

Project Competition: Web Deployment Strategy - Google Gemini

Project Proposal: VivaVox An AI-Powered Dynamic Mock Interview Evaluator

Submitted by: Rushal

Degree: B.Tech, Artificial Intelligence & Data Science

Date: April 2026

1. Project Overview

VivaVox is a cloud-native, multimodal AI web application designed to simulate real-world technical and HR interviews. Moving beyond static, pre-defined questionnaires, this platform parses a student's uploaded resume to dynamically generate highly personalized interview questions. The system conducts a conversational mock interview, analyzes the student's spoken answers, vocal tone, and facial expressions in real-time, and provides a beautifully visualized, comprehensive feedback dashboard.

Competition Focus: Built for instant accessibility and scalability, VivaVox is deployed entirely on the cloud, offering a premium, zero-friction User Interface (UI) and User Experience (UX) that rivals commercial SaaS products.

2. Problem Statement

Students often struggle to transition from academic learning to professional interviews because they lack personalized practice. While standard mock tests provide generic questions, real interviews are heavily based on the candidate's specific resume and projects. Furthermore, traditional text-based tests fail to evaluate crucial soft skills such as confidence, eye contact, and performance under pressure.

3. Proposed Solution

We propose a fully automated, internet-accessible web platform where:

- **Resume-Driven Logic:** The student uploads their resume, which the AI analyzes to extract key skills and projects to formulate a tailored interview path.
- **Dynamic Questioning:** The AI generates tailored questions and natively "speaks" them to the user via the browser.
- **Multimodal Analysis:** The system evaluates video and audio for technical accuracy, confidence, eye contact, and communication clarity.
- **Instant, Visual Feedback:** Results are immediately compiled into an interactive, beautifully designed web dashboard—requiring no external software or downloads.

4. Cloud-Native Technology Stack

To ensure the project is live, fast, and globally available on the internet for the competition jury, we have simplified the architecture to use modern cloud services:

- **Frontend (UI/UX Core):** Vanilla HTML5, CSS3, and modern JavaScript (ES6+). By eliminating heavy frontend frameworks, the application achieves lightning-fast load times and minimal dependencies. Custom CSS will be used to design a premium, accessible UI. Deployed globally on Vercel or GitHub Pages for zero-latency loading.
- **Data Visualization:** Chart.js (a lightweight JavaScript charting library) to build native, beautiful radar and bar charts directly in the app, completely replacing the need for clunky external tools like Power BI.
- **Backend API:** Python FastAPI hosted on Render or Railway. Handles the core NLP scoring and AI prompt orchestration.
- **Database:** Supabase (Managed Cloud PostgreSQL). Provides secure, instant cloud storage for user profiles and interview history.

AI & Machine Learning Pipeline:

- **Client-Side Vision:** MediaPipe Web runs directly in the user's browser via JavaScript to track eye contact and facial expressions. This drastically reduces server costs and ensures zero-latency UI feedback.
- **Resume Parsing & Logic:** Groq/Gemini API free tiers, strictly constrained by Pydantic for structured JSON output.
- **Speech-to-Text:** Groq Whisper API for lightning-fast transcription.

5. Premium UI/UX Design Strategy (Core Focus)

For a competitive presentation, the user experience is paramount. VivaVox is designed with a calming, highly polished "Dark Mode" aesthetic to reduce candidate anxiety and project a premium feel.

- **The Pre-Flight Check (Onboarding):** Before the interview begins, users are guided through a sleek hardware check. Smooth CSS animations and clear green checkmarks confirm microphone, camera, and network status, building user trust immediately.
- **The Interview Arena (Focus Mode):** Cognitive load is minimized. The UI hides real-time analytics and distracting elements. It displays only the current question text, a centered self-view webcam feed (to encourage eye contact), and subtle, pulsing audio visualizers to confirm the system is "listening."
- **Engaging Processing States:** Instead of a static loading spinner while the AI grades the interview, the UI utilizes active status polling via JavaScript (e.g., "Grading Q3/7... Extracting Eye Contact Data...", "Analyzing Vocal Tone..."). This micro-interaction assures the user the app is working and builds anticipation for the results.
- **The Post-Interview Analytics Hub:** The culmination of the experience. The dashboard presents actionable insights using interactive JavaScript charts (Chart.js).
 - **Radar charts** map soft skills (Confidence, Eye Contact, Clarity).

- **Interactive Transcripts** highlight missed technical keywords in red and strong answers in green, allowing the user to read exactly where they went wrong.

6. Simplified Architecture & Continuous Data Flow

By shifting the heavy Computer Vision tasks to the client's browser (MediaPipe Web), the architecture becomes incredibly lean and fast over the internet:

1. **Auth & Upload:** User logs in securely via Supabase Auth and uploads a PDF.
2. **Prompt Generation:** FastAPI backend reads the PDF, queries the LLM, and returns 5-7 customized questions in strict JSON format to the HTML/JS frontend.
3. **The Interview Loop:** The browser natively reads the question aloud. As the user answers, MediaPipe tracks eye contact locally in the browser. Audio is recorded.
4. **Evaluation:** Once an answer is complete, the lightweight audio transcript and the browser-calculated eye-contact scores are sent to the Python backend for final technical/NLP grading.
5. **Dashboard Rendering:** The backend returns the final scores, and JavaScript instantly animates the Chart.js dashboard.

7. Development Timeline & Agile Strategy

A streamlined 8-week timeline focused on shipping a live web product:

- **Phase 1: Cloud Foundation & UI Skeleton (Weeks 1-2):** Set up frontend hosting (HTML/CSS/JS on Vercel) and Render (FastAPI). Connect Supabase. Build the core UI layouts, grid systems, and navigation using custom CSS.
- **Phase 2: AI Logic & PDF Parsing (Weeks 3-4):** Master the LLM prompt engineering to successfully generate questions from resumes. Connect the Web Speech API in JavaScript for reading questions aloud.
- **Phase 3: The Interview Engine (Weeks 5-6):** Implement audio recording and client-side MediaPipe Web for real-time facial/gaze tracking. Ensure the "Continuous Flow" of questions works flawlessly between the vanilla JS client and Python backend.
- **Phase 4: Dashboard & Polish (Weeks 7-8):** Integrate Chart.js. Build the dynamic Post-Interview Analytics Hub. Polish the Pre-Flight check and CSS loading animations. Final bug testing for live internet deployment.

8. Conclusion

VivaVox transforms the mock interview process from a static exercise into an intelligent, adaptive simulation. By utilizing a modern, cloud-native stack and shifting heavy processing to the client side, this project delivers a highly practical, zero-latency web application. Its intense focus on intuitive UI/UX design ensures it doesn't just function well as an AI tool, but stands out as a complete market-ready software product.